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A self-organizational approach to identity and emotions: An overview and implications

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Introduction

The perceptions, ideas, and feelings we hold of ourselves, whether consciously or not, are a natural, intrinsic part of our daily functioning. They seem a necessary ingredient of our attempts to sustain a basic sense of identity, or as Erikson (1950) described it, to protect a sense of personal continuity, unity, and social recognition. As such, they make us an individual in the eyes of others as well as ourselves. But they are also the orientating instruments by which we try to bring some coherence to our own life or to set out new directions. For psychologists, self-referential processes retain an intriguing, though somewhat elusive character, which makes it difficult to capture them in a single, neat conceptual and empirical framework. In a sense, the content of such processes seems to typify the person one has grown to *be*. Yet, in another sense, they constitute the foreshadowing of the person one could *become*. Moreover, in both cases, as James explained, these processes introduce an element of recursivity in the way we experience ourselves, without totally merging with ongoing mental, behavioral and social processes.

In Chapter 1, we stated that in past decades psychologists have tried to deal with the indefinite status of such generic concepts as self and identity by exclusively focusing on the more tangible aspects. This generally meant that self and identity were equated with the self-concept, which was seen as a relatively stable, generalized set of self-representations that people have formed during their lives. This opened the way to the still predominant empirical practice of using self-descriptions and self-ratings as reliable indications of an underlying self-concept. Thus, the more static, decontextualized, and cognitive features were stressed at the expense of the more dynamic, contextual, and emotional qualities.

A general dissatisfaction with the far from spectacular results of traditional self-concept research (e.g., Wylie, 1979) eventually forced self-psychologists to venture in new directions. Van der Meulen's review of recent developments in self-concept theory and research in Chapter 2 provides an extensive discussion of the different attempts to compensate for the lack of attention to affect, context, and variability. She described how the various reconceptualizations and ensuing empirical research to date have shown that the beliefs persons have of themselves are closely associated with affect, and are a mixture of more generalized, enduring self-attributions and more situational and temporal ones. Unfortunately, when taken together, the empirical

findings constitute a rather fragmentary picture of partial aspects, without the promise of an adequately integrated framework. Moreover, these new approaches remain rooted in a view of self and identity as a cognitive, internal structure that analytically and empirically stands apart from the accompanying emotional and contextual processes.

The authors represented in this volume propose a radically new perspective on self and identity, theoretically as well as empirically. The central notion in this approach is the dynamics of self-organization, emerging from the reciprocity of cognitive, emotional, and contextual processes. More specifically, the following key issues are addressed. Firstly, in what way(s) do emotional and cognitive processes implicate each other within the dynamics of self and identity? Secondly, how do these processes in interaction take shape within the contexts in which one is functioning? Thirdly, how do seemingly contradictory characteristics such as stability and variability combine in self-organizing processes? While such issues can be posed in simple and straightforward terms, answering them is a totally different matter. This is reflected in the contributions to this volume. In dealing with the complexity of the dynamics involved in self and identity, each author has developed her or his own vantage point and explanatory devices. Hence, we can imagine that a first reading may have resulted in some confusion about the underlying divergences and similarities between the different contributions. In this final chapter, we will, therefore, start with a brief overview of the chapters in order to identity both the underlying assumptions and important future issues.

A concise overview

Setting the stage for a dynamic perspective

In disputing the exclusively cognitive connotation of self and identity at the outset of this volume, it seemed desirable to approach questions about the role of emotions from the standpoint of emotion theory. This was the task **Frijda** fulfilled in chapter 3. Basic principles in Frijda□s (1986) emotion theory are that emotions indicate what matters to a person, and that they motivate human behavior. One only has to reverse this line of reasoning to come to the conclusion that if self and identity do matter and motivate human behavior they should logically include emotions. This subsequently raises the question in what ways, according to this theory, self and emotions

may be related. In none of the emotion components that Frijda distinguishes is the self a necessary ingredient, at least not in its most common meaning as explicit self-awareness or self-representation. The only prerequisite is what Frijda calls the 'transcendental self': the logical reference point of any experience (including emotions). The transcendental self has no mental content beyond being the locus of a vague sense of familiarity and initiative. In the remainder of his analysis, Frijda actually turns things around. Though aspects of self and identity may not be prerequisites of emotion, it can be shown that, if properly specified, different aspects of self and identity are implicated by emotions.

Firstly, there are many emotions in which the self figures as an object, either in becoming aware of undergoing a particular emotion (e.g. □*I* feel fear□) or as the referent of an emotional appraisal (e.g. 'I hate *myself*' or 'I feel proud of *myself*'). Such self-representations and self-evaluations are often involved in the more adult forms of shame, guilt and pride. Besides, one's own emotions can become objects of emotional appraisal, because of the personal and social implications of having or showing these emotions. This is the 'significance' of an emotion, which motivates the further regulation of emotions. The significance often involves self-representations such as 'a Friesian doesn't cry', or 'I'm an anxious person'.

Secondly, a sense of self may also be considered a possible factor in estimating one's own coping resources. The perceived competence in dealing with the emotion-eliciting event is one of the determinants in the coming about of an emotion, in addition to the appraised event and the individual's concerns. However, such a sense of competence is entirely implicit in the appraisal of the outcome of an emotional event, and no self-representation or conscious self-evaluation needs to be assumed here.

The most pervasive sense in which the self is implicated in emotions is in the form of concerns, of 'what and how the subject is' (Frijda, chapter 3, p. xx). 'Concern' is a broad term for the individual's motives, goals, attachments, aversions, etc., in other words all that matters to the person. Sometimes, the term 'self' is used to denote these personal concerns. Persons can have concerns about themselves, for example the wish to achieve and maintain a certain identity. According to Erikson (1950/1963), the need for a sense of identity is one of the most fundamental human concerns. Frijda (1986) himself mentions self-esteem as one of the most

basic concerns. These concerns, in turn, can lead, in the course of the individual's development, to the more complete forms of self-conscious emotions (e.g. pride, shame).

Summarizing, Frijda's paper provides several pertinent contributions to the central issues in this volume. Though self-representations may not be a necessary ingredient of emotions, emotions do play a major part in self and identity. Most importantly, they indicate what is essential to the person: his or her concerns and competencies. The need for a sense of identity and self-esteem are among the most fundamental human concerns. Moreover, by integrating emotions into the study of self and identity, a direct link between the individual and the context is guaranteed. Because emotions are fundamentally relational, fusing the subject and the object of an emotion, this integration can resolve the problems inherent in the decontextualized nature of self-representations research. The conceptualization of emotion as a system of loosely coupled affective, cognitive and behavioral components (see also Frijda, 1993) offers new opportunities for an understanding of how emotions and self may be intertwined. It also preludes a dynamic systems perspective, because the mutual interactions between the different emotion components and aspects of self are the defining elements in the eventual course of the emotion processes.

In his critical commentary, Epstein elaborates on the relation between emotions and self-aspects in a way that to an important extent goes along with Frijda's componential and dynamic view. However, he relegates much of Frijda's emotion theory to the so-called 'experiential system'. He describes the experiential system as the executive part of the self, and as something that has to be clearly separated from the self as the object of rational considerations. The system operates on the basis of an intuitive theory about oneself, about the world, and about their interrelationship. In this system, emotions indicate what is of significance to a person. Frijda and Epstein also agree that the dynamic interactions between implicit appraisals and emotions are largely non-conscious, but both describe the dynamics involved only in global terms. In this connection, the application of dynamic systems theory might be very useful.

Chapter 4 by **Van Geert** is intended to be a general introduction to dynamic systems thinking. Since many contributors to this volume use or advocate a dynamic systems perspective on issues of self, identity, emotions, and development, this chapter offers the necessary background information for a preliminary evaluation of the usefulness of such a perspective. A dynamic

systems perspective offers insight into processes, and thereby into questions concerning stability and change. Van Geert describes a dynamic system as something that 'consists of elements that exert specific influences or forces upon one another and by doing so, change each other's and their own properties' (chapter 4, p. xx). In their commentary, Camras and Michel correctly point to one of the critical implications of this definition. In applying a dynamic systems perspective, one should define constructs in terms of systems, that is, in terms of a conglomerate of elements (e.g., prey and predator fish). In this volume, therefore, the question has to be posed to what conglomerate of elements the generic terms 'identity' or 'self' refer. It should also become clear what is part of the system and what is part of the system's environment. Indeed, most authors in this volume have in one way or another, implicitly or explicitly, tried to deal with these implications.

An important view of dynamic system theory is that in order to explain variation over time, that is patterns of stability and change, it is not necessary to resort to external causal forces, or an inbuilt, deterministic ground plan. Instead, the most conspicuous characteristic of dynamic systems is self-organization. Complex, non-linear systems tend to self-organize into a state of increased order/complexity. Higher order phenomena, for instance self-constructs or identity, can be seen as emerging from the dynamics of the system rather than being already present as internal structures. In dynamic systems terms, changes result from the time-dependent interactions between the constituent components of the system. These interactions -- the set of relationships between the components of the system -- form the core of a dynamic systems model, and not the physical instantiation of the components (e.g., molecules, fish, classroom reprimands, self-representations, identity choices). This means that the general process characteristics of dynamic systems, such as different types of attractors and bifurcations, can be applied to describe the dynamics in a specific domain such as self and identity. At a qualitative level, this can have an important heuristic value, but, as Van Geert has demonstrated, the relationships between the components can also be specified in mathematical functions which are surprisingly simple but which nevertheless capture the essence of the system's behavior over time.

Four elaborations on a theme

The chapter by Frijda and by Van Geert have informed us about the role of self-phenomena in emotions, and about the application of a dynamic systems perspective in order to approach questions concerning stability and change. Chapters 5 through 8 provide in-depth discussions of emotions and context as formative conditions from which self and identity emerge in a self-organizational process. However, each of these chapters has a different focus. Nevertheless, in certain ways, they also complement each other and build on the foundations laid by Frijda and Van Geert.

Frijda makes it clear that one's self can become the object of emotions. In order to experience oneself as an object of an emotion, some notion of oneself as enduring and stable over time must exist. But, when and how, in ontogeny, do these self-related phenomena emerge? These are the kinds of questions **Fogel** explored in Chapter 5.

Fogel based his approach on two premises: Psychological experience (a) always implies a relationship (with the object-world, others, or oneself) and (b) is always dynamic and changing. Within this inherently relational and dynamic world, the crucial question then becomes how people can ever develop a sense of uniqueness and continuity [1]. Fogel considers emotions as playing a central role in this process, because they are a way of discovering the meaning of a relationship for the self and, hence, the unique position of the self in a relationship. The perception of consistency in one's emotional experience begins to yield a sense of permanence of the self over time. With these premises and his almost programmatic statements, Fogel occupies a distinctive position in the context of this volume: All experience is dynamic, self is rooted in relationships, and emotions (which in themselves are inherently contextual) play a constitutive role.

Central to Fogel's analysis of the dynamics of self and identity is his distinction between three time scales, and the observation that, depending on the time scale, different types of emotions go together with different experiences of uniqueness. A first and basic sense of self already comes

1. Fogel's question is a direct consequence of his relational and dynamic perspective. Authors working from an individualistic perspective, on the other hand, tend to posit stability and uniqueness and have to go to great lengths to conceptualize change and connectedness (e.g., Grotevant & Cooper, 1986; Josselson, 1994).

into existence when the emotional experience that accompanies a currently ongoing relationship stabilizes into a primary orientation towards the relationship. This takes place on the time scale of (micro-) seconds. In turn, orientations coalesce into identifiable events. On the time scale of minutes to days, sequences of events may self-organize into coherent and stable frames. Frames are similar to narrative scripts and enable individuals to experience a more explicit, stable sense of self. The third time scale, the scale of 'development', extends over years. The corresponding sense of self is identity: the experience of an enduring self. It is rooted in the relation with one's past or future, but also in the surrounding cultural frames. Although identity may develop into a narrative account of oneself, Fogel considers it to be an emotional experience rather than a representation.

Regarding the main questions considered in this volume, Frijda and Fogel have much in common. Both argue for a differentiated conceptualization of self and identity, and both agree that emotions are relational experiences. In a certain sense, they also complement each other. Frijda makes clear that self-related phenomena are not a necessary part of the emotional experience. Fogel gives a similar primacy to emotions but seeks to understand how aspects of the sense of self emerge from emotional experience in relationships. In our opinion, Fogel's description of the emergence of a sense of self on three different but interrelated time scales offers a significant and elegant contribution to our understanding of the constitutive role of emotions in the development of aspects of self. It is interesting to note that Fogel hardly refers to explicit representational content. Apparently, he, like Frijda, considers self-representations and self-awareness to be unnecessary in understanding the way in which the self becomes organized.

In his commentary on Fogel, Jansz suggests an alternative explanation for the emergence of a sense of identity. In contrast to Fogel, he sees identity and its constituent feelings of continuity and consistency mainly as a social construction that people have learned to ascribe to themselves. In this they are guided by the prevailing notion of personhood, which in Western societies is characterized by an emphasis on personal consistency, uniqueness, autonomy, and rationality. In his reply to Jansz (not published) Fogel states that the difference between his and Jansz'view stems from a difference in starting point. In Fogel's view structures have to be explained, whereas relations need no explanation. Jansz starts from structures and has to explain relations.

Thus, Jansz emphasizes the role of cultural scripts in formatting identity through dialogical exchange, and the cultural fact that emotions count as a violation of the aspired image of equanimity. Fogel, on the other hand, concentrates on the stabilizing role of emotions in the emergence of a clear sense of identity within dialogical relationships Both ways of relating emotions to the dialogical nature of identity return in Chapter 6. **Hermans and Hermans-Jansen** combine Fogel's relational perspective with culturally determined patterns of meaning-making by introducing a new element: a strong recognition of the developmental possibilities that are brought about by the human faculty for self-reflection. They start from the assumption that in the course of human development emotional responses are increasingly influenced by the relationship individuals have with themselves. In terms of the differentiation Frijda makes, Hermans and Hermans-Jansen focus on experiences in which the subject becomes the object of the emotion, and in particular on the significance of the emotion. The significance plays an important role in emotion regulation, in the way individuals interpret and express, or do not express, their emotions. This is also the domain in which the culturally determined emotion scripts figure.

In their analysis of the dynamics and complexity of the relation between self and affect, Hermans and Hermans-Jansen elaborate on James' distinction between 'I', i.e., the self-as-knower, and 'Me', i.e., the self-as-known. They see self-awareness as an essential feature of the I. Individuals can reflect on their feelings, appraise them and develop feelings about these feelings. Thus, the self authors dynamic and complex compositions of personally significant, affect-laden meanings (i.e., valuations). Such narratives typically take the shape of an ongoing self-dialogue. Instead of being the omnipotent author of oneself, the self-reflecting individual continuously changes perspectives. Each perspective in fact implies a different I-position, entailing its own internalized voice and telling its own story. The different narrative positions usually revolve around recurring conflicts and obstacles one encounters in becoming the person one wants to be or should be. Thus, self-narratives have developmental implications. To serve as a fruitful point of departure for further identity development, the inner voices that speak for the own person should be brought into open dialogue with each other.

The notion of the 'dialogical self' returns in the valuation theory and the self-confrontation method, which can be employed to study the dialogical self empirically. Both in its conceptual connotations and its methodology, the approach of the dialogical self has much in common with the phenomenological-dialectical personality model of Verhofstadt-Denève, that is used as starting point in the application of psychodrama techniques.

The authors of Chapter 7**, Haviland-Jones, Boulifard and Magai**, present a similar componential conceptualization of the self. By the self they mean self-concept, identity, or personality, terms which they use more or less synonymously. They take emotions as a starting point and study how emotions organize personality, in contrast to Hermans and Hermans-Jansen, who argue that I-positions organize patterns of emotions. Haviland-Jones and her colleagues give a lively account of their quest for tools that enable them to arrive at a better understanding of the dynamic interplay of cognitive and affective components of the self. They suggest that dynamic systems thinking and modeling offer exciting new possibilities. Their earlier studies failed to provide adequate insight into the role of emotions in processes of developmental change in self-concept and identity. It was their idea, therefore, that analyzing *one* aspect of these developmental processes might provide further insight. This brought Haviland-Jones and colleagues to the study of micro-processes of fear by means of a dynamic systems model. They base their model of emotion on the theory of Tomkins. In the model they combine the idea of continuous change in the system's input with various discrete emotions as its categorical output. Such a model also allows for continuous transitions from one emotion to another. The dynamic systems methodology developed by Van Geert makes it possible to run computer simulations of these continuous emotional processes and to study the complex interactions involved.

The computer simulations of emotion processes carried out by Haviland-Jones, Boulifard, and Magai suggest that differences in the original state of the person, or in the stimulus itself may make large differences in the emotional experience, with implications for immediate behavior as well as long-term identity scripts. Haviland-Jones and colleagues have demonstrated that information parameters are in themselves capable of producing shifts in emotion categories. Furthermore, conscious, goal-oriented knowledge does not seem to be essential for these shifts.

The model and simulations also give room to interesting speculations about the interdependencies of emotional and cognitive processes: Brief and unconscious emotional processes could evidence 'phase shifts', which could even lead to changes in stable patterns in the person's self-concept. Beliefs and knowledge about emotions also play a role in the processing of self-relevant information. They may enable individuals to change themselves, but due to the intricacies of the mostly unconscious dynamics of micro-emotional processes, such changes can take unpredictable courses. These may or may not correspond with how individuals understand and describe themselves: A 'good linear story' about how one came to be -- call it one's narrative or one's meaning making -- does not necessarily reflect the workings of the micro-emotional processes in the change and stability of the self. Haviland-Jones and colleagues conclude by suggesting that the iteration of the phase shifts on the micro-level and the meaning making level may lead to changes in one's self-view.

In his commentary on Chapter 7 Heymans does not contest this suggestion, but he warns us that one should not forget the role of cultural constructions of the self in the individual's developmental process, or the individual's own active involvement. In line with this, he also observes a conceptual gap between the description of self-organizational processes occurring at the micro-level of ongoing, mostly unconscious emotion processes, and at the macro-level of active, conscious identity construction.

Heymans broaches an important and thorny issue that is still left unanswered. Throughout this volume, there seems to be a dichotomy in the kind of explanations that are used to describe the dynamic organization of self and identity. Self and identity are either seen as emergent qualities of emotion-driven self-organizational processes (Fogel, Haviland-Jones et al.) or as a self-reflective construction of meanings (Hermans & Hermans-Jansen, Jansz, Heymans).

In Chapter 8, **Lewis and Ferrari** attempt to unite these two types of explanation under the header of 'implicit' versus 'explicit identity'. Unlike Epstein's CEST, Lewis and Ferrari do not try to solve this issue by postulating two totally independent self-systems. Instead, they strictly adhere to the organizing principles of dynamic systems, principles that help to explain how complex, higher-order organization and stable states emerge from the interactions between

lower-order components. As such, Lewis and Ferrari present an encompassing application of dynamic systems thinking in the domain of self and identity.

According to Lewis and Ferrari, implicit identity -- the sameness and uniqueness of our personality -- is a naturally evolving characteristic that we have in common with most animals and that already exists in young infants. It emerges from recurrent, self-stabilizing patterning of appraisal-emotion interactions. The self-organizational processes involved build on each other along three different time scales. In real time, the appraisals and emotional states that accompany our actions are coupled into temporary 'emotional interpretations' (EIs). These EIs dissipate as soon as the initial goals are achieved. When we are unable to achieve the goals, the goal state perseveres. The emotions that this situation generates have a consolidating effect on the current EIs, turning them into more or less lasting moods that may persist for minutes to weeks. Such moods have a facilitating effect on the interpretations and action-tendencies that are congruent with them. They also constrain the likelihood that alternative EIs develop into a mood. Through this habitual shaping of the mind, the configuration of EIs that fit in with a recurrent mood may start to function as a psychological attractor state, and become a more permanent ingredient of our personality. This is especially the case when long-lasting needs and wishes are involved that are difficult to accomplish, given the personal circumstances one lives in. In a nutshell, then, psychological continuity crystallizes from the mind's preoccupation with 'unfinished business', rather than being the intrinsic property of an underlying, invariant self-structure.

A fully developed personal identity relies on the sameness of the implicit identity, but also implies a coherent composition of self-appraisals. This is what Lewis and Ferrari call the 'explicit identity'. It contains people's 'statements, stances, policies, or stories about who they are and how they live their lives' (p. xx), and stems from semantic, reflective, and often conscious activities of a more autobiographical and dialogical nature. Again, such self-reflective activities occur especially when people have to deal with unresolved needs and goals or the conflicts that this arouses. The main function of self-narratives and self-dialogues is to account for the fact that one has not yet succeeded in becoming the person one wants to be. This is done in anticipation of real or imagined listeners. However, since the potential audience can differ

widely in their appreciation of the narrator's intentions, and the narrating individual can adopt a multitude of I-positions, there can be no conclusive understanding of oneself.

Lewis and Ferrari also stress that from a dynamic systems perspective, identity change can be modeled through different kinds of change in self-organizing systems. Identity change across developmental time may stem from the emergence of new patterns, the strengthening of new habits, and dramatic events. However, in her commentary, Magai tries to qualify Lewis and Ferrari's assertion that the same dynamic system principles govern change and stability in implicit and explicit identity. She refers to the empirical evidence that, precisely because of its dialogical nature, explicit identity appears to be much more susceptible to change than the more deeply entrenched implicit identity. She therefore wonders whether at a more specific level each type of identity may imply rather different types of organizing mechanisms and processes. This is one of the fundamental questions that will be discussed in the following section. In their reaction (Lewis & Ferrari, unpublished) agree that there are important differences between identity and personality: Not only does identity change more easily, but probably it can also change more partially. They suggest however that the differences between both systems do not reside in the general dynamic systems principles, but in the more specific mechanisms and processes on which these general principles operate.

In the course of Lewis and Ferrari's comprehensive discussion, many of the conclusions of previous chapters reappear: the emphasis of Frijda, Fogel, and Haviland-Jones and her colleagues on the organizing role of emotions in self and identity; Van Geert's outline of dynamic system principles; Fogel's distinction between three time scales and levels of organization; and Hermans and Hermans-Jansen's sketch of the dialogical self. However, as Magai already observed, this comprehensiveness has also carries the cost of addressing the dynamics of self and identity in a general and abstract way. In their study of these dynamics, researchers need to be aware that there are some decisive analytical implications that have to be taken into account when the general principles of self-organization are translated to the domain of self and identity. We will, therefore, conclude this closing chapter by discussing some of the most compelling of these implications.

Implications of a dynamic systems approach to self and identity

In the first two chapters of this book three broad problem areas were identified in recent developments in the study of self and identity: the role of emotions in a predominantly cognitivistic approach, the question of how to take the context into account, and the issue of stability versus variability and change. Most of the authors in this book share a perspective that differs fundamentally from the approaches that have to contend with these three problems. They see emotions and context as formative conditions from which self and identity emerge in a self-organizational process. In such a dynamic systems perspective the various elements ‘ cognitions, emotions, and context ‘ are viewed as a complex, interacting network in which the embeddedness in the context and the changeability are inherently given. The limitations of the traditional cognitivistic approach can be overcome via this new perspective.

The use of a dynamic systems perspective has brought us a wealth of new ideas, but it also raises new questions; the ensuing implications primarily have to do with how the of under study is approached and defined.

In his chapter Van Geert distinguishes four steps necessary for applying such a perspective. The first step, conceptual construction, concerns the definition of the components of the system. In addition, Camras and Michel stress the point that a dynamic system always consists of multiple layers, and that the phenomena emerging from these components should also be defined clearly. Further, it must also be clear what exactly is meant by the different concepts used in the self and identity literature (e.g., self-concept versus identity). A first implication therefore concerns an elaboration of self and identity in terms of multiple layers of components and emerging phenomena. As a further implication we will discuss the distinction between the various concepts that are central to this volume.

The second step in Van Geert’s description concerns the formulation of the interactions between the components. Applying a self-organizational perspective to self and identity implies that the processes involved in the interactions between the components, within and between levels, have to be described. This is another implication to be elaborated here. Furthermore, as Camras and

Michel point out, the relation between identity development and its context has also to be defined. Yet another implication is that there are alternative ways of conceptualizing this relation.

The third step mentioned by Van Geert concerns a quantitative approach: the translation of the conceptual model into a mathematical one. Elaboration of this step goes beyond the scope of the present chapter, although we will, by way of an example, briefly describe how different conceptualizations of the context could be translated into a mathematical model.

Van Geert's final step concerns the exploration of the model. This, of course, is needed both in a qualitative and in a quantitative approach: Researchers have to formulate hypotheses and test these against empirical data. A final implication of a dynamic-systems approach has to do with the kind of data and methods of analysis that are needed for testing such hypotheses.

Components and emerging characteristics

A dynamic system is - by definition - multilayered. This means that components and emerging characteristics should be defined separately for each level of aggregation: A phenomenon that is an emerging characteristic on one level can be a component on a higher level. Although the hierarchies and components discussed by the various authors are not always comparable, we will elaborate on how different levels of organization can be distinguished and ordered with regard to self and identity; we will then try to relate the concepts discussed in this book to these different levels.

The lowest level (see Table 9.1.) discussed in this volume consists of components such as feelings, bodily sensations, cognitions, orientations, perceptions, appraisals, action tendencies, etc. This is a somewhat arbitrary choice of 'lowest level', because these components, for example bodily sensations, can also be seen as higher-order phenomena that emerge from components of an even lower order. Coupling of these components results at the second level in the emergence of (higher-order) temporary emotional interpretations, which are probably comparable - in terms of level of aggregation - with what Fogel calls events. On the third level these events self-organize over time into enduring phenomena: recurrent emotional interpretations or Emotional Interpretations (Lewis & Ferrari), frames (Fogel), roles (Haviland-Jones et al.), and valuations (Hermans and Hermans-Jansen). Self-evaluations and self-representations can also be situated on this level.

On the fourth level, self-organizing EI's give rise to implicit and explicit identity (Lewis and Ferrari).'The self one is' (Frijda), sense of self, sense of identity (Fogel) and self-theory (Epstein) can also be situated on this fourth level of organization. (For reasons that will be explained below, self-concept is not included in the table.)

Insert table 9.1 about here

Table 9.1. provides an overview of how - from a self-organizational perspective - different concepts can be related to each other. The table is not intended to be the basis of an exhaustive definition of the one and only system of self and identity. The definition and the defining components that are used depend on the concept on which a particular study focuses. Hermans and Hermans-Jansen, for example, focus on the narrative organization of valuations in their case study. Each valuation can be seen as a system, consisting of events, temporary emotional interpretations, emotions and cognitions to begin with, but they become indicative of self and identity because they cohere into a meaningful whole. Haviland-Jones and colleagues, on the other hand, describe a system that consists of lower-order components and emerging patterns of emotions.

The various levels are related to different time scales. Lowest-level components self-organize in events, that is, momentary emotional experiences on a microscopic time scale; experiences subsequently organize in EI's on a medium time scale; and identity, finally, emerges on the highest time scale. The role of emotions and the relation with the context differs across the different time scales. It is essential, therefore, to be clear about the time scale one is talking about. From a systems perspective it is also important to realize that the levels of self-organization presented here are in turn part of a much more extensive system of systems. Fogel, for example, sees phenomena like history and culture as constructs of an even higher order.

Conceptualizing self and identity as emerging from a multilayered system of interacting components has some major implications. Firstly, the question 'Which comes first, emotions or cognitions?' becomes a meaningless one. There is a continuous interaction between all components, EI's themselves being a conglomeration of emotions, perceptions, cognitions, etc.

Secondly, this conceptualization implies that constructs like self-concept, self-esteem, self-evaluations, and all kinds of ideal, real, feared and other selves, are entities that do not pre-exist in a person's mind. Stability does not mean that the construct is always there (in this sense the word construct is inadequate, because it suggests something that is always there), but that comparable phenomena emerge in different situations over time. For example, a similar self-evaluation may emerge during many years in different situations in which an individual's attention to his or her own person is drawn. What situations they are depends on the person, the moment and the situation. Since self-related phenomena self-organize at specific moments, they will be comparable, but they will probably also vary systematically in different situations. A self-organizational perspective does not necessarily require the defining of different kinds of selves ('self at school', 'self as I would like to be', etc.) to account for these systematic differences. It is therefore much more economical than essentialistic theories, which do need such a priori selves. Moreover, since emotions and context are fully present in the system at the most basic level of self-organization, there is no need to add them as separate constructs later on.

Sense of identity, implicit and explicit identity

We have so far argued that in dealing with self and identity it is essential to make a distinction between different levels of aggregation. Processes on all levels play an important role but studies of self and identity usually focus on long-term developmental processes, that is, concepts on the third and fourth level. In general, the concepts mentioned on the third level are clearly defined throughout the book. Concepts and definitions on the fourth level, however, cause considerable confusion. The concepts on this level are all global ones, they concern the person as a whole, and emerge from the same underlying layers. What they have in common, and what distinguishes them from other constructs in psychology, for example 'the person' and 'personality', is that they refer to the person's own perspective upon and experience of him- or herself.

'Self' and 'identity' are the most encompassing constructs in Table 9.1. As stated in Chapter 1, we treat them as interchangeable. We prefer to continue with the term 'identity' only because of its more circumscribed meaning. The differentiation made by Lewis and Ferrari between 'implicit' and 'explicit' identity is especially helpful. Apart from 'a sense of identity', all fourth-order constructs in Table 9.1. are covered by these two constructs. Thus, sense of identity,

implicit identity and explicit identity will be the three central constructs used in the remainder of this chapter.

Sense of identity is a general feeling, a sense of 'sameness and continuity'(Erikson, 1969; see also Blasi and Glodis, 1995) that runs through the whole system. Because of its coordinating character, it should be situated on the highest level of abstraction (in Table 9.1). The emergence of a sense of identity has nothing to do with the content of any of the components, but everything to do with the way the different components fit together over time. Continuity of emotions, of frames, and of emotional interpretations gives rise to a feeling of being the same over time. Thus, a sense of identity may emerge from all three lower levels of aggregation: the underlying system of components, self-organized events and EI's.

Paradox is inherent in a sense of identity: If people have a sense of identity, they are (generally) not aware of it (Erikson, 1969). It manifests itself mainly as a sense of well-being; from a self-organizational perspective this means having a total absence of any positive feedback loops in the system. Positive feedback loops fixate, stabilize. In their absence, the system can be seen as a completely free-floating continuous relationship in which all kinds of change are possible, as in the state that Fogel describes as purely orientational. This conceptualization underlines the fact that a sense of identity refers to a fit and is fundamentally relational. The absence of a sense of identity, on the other hand, may be felt keenly and consciously, and may give rise to heightened and pre-occupied attention to identity issues (Erikson, 1969).

Implicit identity is the organization of especially non-conscious EI's a person has of him- or herself. It resembles Epstein's concept of 'implicit self-theory', which - together with a 'world theory' and notions about the relations between these two – makes up the experiential system. This experiential system is comparable to what Lewis and Ferrari call personality. As Epstein points out, there is no sharp boundary between a person's self-theory and world theory. In fact, they are two sides of the same coin. For example, the notion that 'the world is a dangerous place' implies that the person sees him- or herself as vulnerable. Nevertheless, the chance that mutual couplings will emerge is higher where both EI's concern oneself. This will result in the formation of a subsystem, the internal connections of which are much stronger than the

connections with components that do not belong to the subsystem. Implicit identity can be distinguished, therefore, as a self-organized system within personality.

This implies that specific EI's can be more or less categorized in terms of whether they are part of identity. Most components on the lower levels cannot in themselves be categorized as being or not being part of implicit identity. A feeling may be part of an EI concerning oneself, but also of some EI concerning the world. Frijda distinguishes several aspects of emotions that have to do with self: Emotions that have oneself as object, the significance of the emotion, implicit estimates about one's own competence, and the two basic concerns of self-esteem and sense of identity. We assume that at least one such aspect must be present before an EI concerning oneself can possibly emerge. However, all elements that are part of personality may become part of the implicit identity.

Explicit identity is an organization of conscious EI's about oneself. It is auto-biographical and dialogical. It resembles the concept of narrative (Hermans and Hermans-Jansen), and explicit self-theory (Epstein). Explicit identity and related concepts have been clearly described elsewhere in this book, and need no further explanation here. As compared with implicit identity, explicit identity is organized in a more compartmentalized way, which means that parts may change, while the rest remains the same.

Other concepts in the field differ from self and identity mainly with regard to their inclusiveness. Camras and Michel, for example, suggest that self-concept could be conceived as a component of identity viewed as a dynamic system. We prefer to view self-representations as components of identity. Self-concept could then be defined as the collection of descriptions of 'who one is', 'who one likes to be', 'how one thinks others see oneself', etc., and therefore as conscious self-representations that are part of one's identity. In this view self-concept is not an emerging higher-order construct: as a collection it is no more than the sum of its components. Consequently, self-evaluations are defined as those self-representations that include emotions, values and judgements about oneself.

Apart from the differences in content described above and elsewhere in the book, differentiating the concept of identity in implicit identity, explicit identity, and sense of identity is also

necessary because, as will be argued below, the processes involved and the assessment procedures differ according to which construct is entailed.

Processes

Now that identity has been described as a self-organizing system, we can examine the processes involved. Both implicit and explicit identity emerge from underlying levels. Lewis and Ferrari suggest that the mechanisms involved might be comparable. However, as Magai suggests, the processes involved in the emergence of implicit identity might well differ from those playing a role in conscious phenomena like explicit identity. She points out that explicit and implicit identity differ in susceptibility to external influences and in probability of change. This would mean that there are different processes involved in the two systems, as is also argued by Epstein and Pacini (1999).

Camras and Michel's distinction between verbal explanation versus formal mathematical models is relevant in this controversy. It may be that the differences are restricted to the verbal model. The formal causes and types of mechanisms can be the same in systems that differ strongly with regard to the verbal causes and variables involved.

We will first consider the processes within the two identities on a verbal-psychological level. Various theories suggest that these are quite different. Epstein states that the processes in the experiential system are based on non-conscious, rapid association, and those in the rational system on deduction and reasoning. In a similar way, Dannefer and Perlmutter (1990) distinguish between two different processes that coordinate between inner-biological, individual-psychological, cultural-sociological, and outer-physical processes,; such a distinction can also be related to the two types of identity. The first process, habituation, is a more-or-less automatic, self-regulatory process that is characterized by routine, non-reflective interaction with the environment. This process contributes to the non-conscious maintenance of a sense of identity. It works on the level of emotional interpretations and includes all kinds of biased types of information processing that prevent people from becoming aware of discrepancies with the desired perception of themselves. This is a discrepancy-reducing, equilibrium-seeking process. The second process, cognitive generativity, is an intentional, reflective form of coordination: the ability to take oneself as object of thought, to reflect upon the past and the present, to anticipate

the future, and to speculate about the unknown. This resembles internal dialogues as described by Hermans and Hermans-Jansen, and explicit identity or narratives as described by Lewis and Ferrari. Cognitive generativity is a conscious attempt to modify and mobilize habitual processes that have become maladaptive. The process works on the level of the deliberate change of one's narrative and is based on the rules of communication and narratives. One of the consequences of this may be that explicit identity can change in a more compartmentalized way (Lewis & Ferrari, unpublished). One of the commitments or voices in the dialogue may weaken, disappear, and be replaced without necessarily turning the whole system upside down.

It is not clear how the processes in the two systems are related. Do they affect each other? Are they active alternately, or do they run in parallel? Epstein describes how the rational system may 'correct' the experiential system if reflection and conscious reasoning 'detect' maladaptive thoughts or actions. The experiential system, in turn, may affect or bias the rational system in a non-conscious way. People do not generally reflect upon themselves most of the time, however. Cognitive generativity might be activated following the inability of the habituation process to dissipate emotions (compare Jansz). This may disrupt one's sense of identity and result in the emergence of strong emotions. Such strong emotions may trigger self-reflection, as Oosterwegel suggests.

On a macroscopic time scale, we see that the developmental courses of the two types of identity differ strongly. According to Erikson (1968), early characteristics of implicit identity are present from birth. One's explicit identity becomes an imperative from adolescence onwards. The person is then confronted with a combination of biological, social, and cognitive changes that makes his or her implicit identity insufficient for safeguarding a sense of identity, resulting in feelings of identity confusion. People will then try to retrieve their sense of identity, and they do so by constructing a personal identity: a tentative, reflectively construed configuration of self-descriptions that allows the person to find his or her own place in the world. A new qualitative dimension is thereby added to the system. This new dimension does not differ from the sense of identity in terms of level of aggregation, but in terms of level of consciousness. Its emergence is the beginning of an ongoing dynamic process of identity maintenance: a more deliberate process of stabilization and change. Every time that - due to developmental growth, changes in situations

or in underlying goals and needs - feelings of identity confusion occur, active and conscious changes in explicit identity may follow. Such changes in explicit identity probably have consequences for implicit identity. Identity development can thus be seen as an ongoing interaction between implicit and explicit identity, between non-conscious and conscious processes.

The case studies discussed by Hermans and Hermans-Jansen, and by Verhofstadt-Denève and the intervention methods used in these cases provide some additional suggestions about the how and when of developmental switches and of the dynamics involved insuch changes. Both of the clients involved sought help in a period of distress. Initially, the reason for this distress was not part of their conscious, explicit identity. During the therapy, different parts of the non-conscious system were brought into awareness. Especially in the case of Sharon, the client's troubles seem to stem from the existence of two different organizations, or attractors, and from a lack of communication between them. The aim of the therapy is to bring the two attractors together, by making the person aware of their existence, and by stimulating the person to communicate between the two perspectives. From a self-organizational perspective, one can describe both therapies as attempts to lower the divisions between different 'basins' of attraction, by coupling components of the two attractors in new and different ways, as attempts to build a new self-organizing attractor, new EI's. In Verhofstadt-Denève's psychodrama, one of these attractors or voices is as it were 'moved' to another person. Distress, in general, may result in a focus on what is going on, in an increasing awareness of previously non-conscious EI's. This awareness enables a deliberate change of the couplings and thus the EI's. The non-conscious system is then changed via conscious intervention. This is not to say that all couplings can be changed at will. Haviland-Jones and colleagues have argued that the construction and change of one's own identity may also result from unconscious micro-level processes and have consequences other than those expected.

Although the existence of different attractors in the self-system can be problematic, people in general have several EI's. This raises the question of when the existence of different attractors becomes troublesome. One could imagine that conflicts between EI's arise if they are both triggered in similar sorts of situations, in which they fulfil different needs, or if they threaten

each other's needs. For example, an interpretation of oneself as being strong and independent may threaten the need for belonging. Epstein (1998b) has described how different conflicts and dis-balances between needs can result in distress.

From the foregoing, it may be obvious that on a verbal level of understanding the processes involved in implicit and explicit identity clearly differ. This does not tell us anything, however, about the differences from a formal, dynamic systems perspective. From this perspective, self and identity comprise multiple levels of aggregation, with two types of self-organizational processes: within one level of aggregation (horizontal) and between different levels (vertical: bottom-up and top-down). These processes take place on and between all levels of aggregation. Lewis and Ferrari introduce two basic principles of self-organization, which together explain how lower-order components give rise to stable higher-order phenomena (bottom-up processes). Firstly, components couple, and this coupling results in the emergence of higher-order forms; and secondly, subsequent recurrent coupling may lead to the stabilization and strengthening of these forms. Crucial in their approach is the idea that recurrence takes place when experiences cannot be brought to a completely satisfyingcompletion, which leads to unresolved intentions and blocked-off actions. Depending on the strength of the accompanying emotions, such experiences may form the basis of recurrent and salient attractors. Mechanisms like these make it possible to explain why some experiences develop into recurrent, salient, characteristic patterns and others do not. Another elaboration of bottom-up processes can be found in Nowak, Vallacher, Tesser and Borkowski, (2000). They demonstrate, by means of a cellular automaton, how interacting elements give rise to higher order phenomena such as the global evaluation of the self. The driving force in this model is the press towards integration, i.e., the striving for coherence between different components in the self-concept.

In contrast to the emergence of higher-order phenomena from the interaction of lower-order components, other processes work top-down: Higher-order phenomena may affect lower-order phenomena. In his reply, Van Geert (unpublished) has discussed two possible forms of these top-down influences. Firstly, newly emerging phenomena may affect streams of energy or information in the lower-order system. One's narrative may affect the way one appraises new situations. For example, the appraisal of computer-related problems will change if one starts to

see oneself as a computer specialist. People may in a conscious and deliberate way change their behavior and environment and try to adjust their interpretations, feelings and thoughts to their own narrative. Secondly, a new phenomenon may be of the same order as the components from which it emerges, and become part of the system. A newly emerging role, for example becoming a mother, will lead to new couplings and may destroy old ones. In this way, the new role affects other roles.

A process that occurs within one level (horizontal) is the so-called phase transition: a change in the organization of the higher-order phenomenon, which can be triggered by small changes in underlying, control variables. Sudden changes may depend on gradual changes in control parameters in a very non-linear way, and their occurrence often depends on all kinds of system characteristics. This means that a transition never has just one cause, but takes place as a consequence of the coincidence of many interacting variables: the state of the whole system and its context. Moreover, major and sudden changes do not need major external events for their occurrence, but can be caused by gradual, and possibly invisible, processes within the system itself.

In physical systems, self-organization is assumed to be activated by a striving for the lowest level of energy. Lewis and Ferrari assume that psychological systems may be activated (or motivated) by the striving for a smooth stream of information or energy - flow in Csikszentmihalyi's (1990) terms. In flow, the stream of information or energy is not hampered, does not get stuck in recurrent patterns, but is free floating. Several authors in this volume assume the existence of so-called basic needs that activate development. Frustration of these needs might block the stream of energy and disrupt the sense of identity. On a non-conscious level, this may result in the formation of new EI's. On a conscious level, in their narrative identity people aim at stability, continuity, and coherence. A stable, enduring and coherent narrative enables rapid processing and integration of information. A disruption of one's narrative means that there is new information that cannot be processed and integrated. This calls for reflective forms of coordination to change the narrative in such a way that the new information can be integrated.

At this moment there is little to say about the difference between implicit and explicit identity as represented in formal (mathematical) models. Little is known about the formal characteristics of

the mechanisms we described above in verbal terms. In their reaction to Magai, Lewis and Ferrari (unpublished) refer to Allan Schore who, in a personal communication (August, 1999), suggested that implicit and explicit identity may rely on altogether different memory systems, yet both develop in parallel and interact continually. For one thing, we have no idea about the developmental trajectories of either system, which leaves us with several questions. Do changes in the two systems always have the form of sudden transitions from one attractor to another, as Lewis and Ferrari describe? Probably, gradual changes are also possible. Are there comparable process parameters for the two systems? What does it mean on a formal level if underlying memory systems differ? Implicit identity aims at equilibrium, but is equilibrium also important for explicit identity? Our guess is that equilibrium is not especially important in the case of explicit identity. A narrative should be convincing, but does not need to be consistent. A good narrative even requires some degree of inconsistency and dramatic tension (Bruner, 1990). In Lewis and Ferrari's opinion retelling a narrative over and over again strengthens it in the same way as happens with repeated couplings on the implicit level. Are change processes within the two systems comparable? This is hard to determine. To be able to answer the question of differences on the formal level, much more needs to be known about the formal characteristics of the emergence and development of self and identity.

Systems and their context

Van Geert states that in a dynamic systems approach the context may refer to the whole universe, with the exception of the system itself. However, it is far from economical to include the universe in one's model. Instead, it is the task of the researcher to define what aspects of the context are expected to be relevant for the system under study. In this regard, the authors in the present volume are notof one opinion. Fogel's approach is fundamentally relational: The focus of his study is on the characteristics of relations rather than on those of the person or the context. Haviland-Jones and colleagues have a different approach. In their view, the context is important in another way. First, it offers specific situations, such as a prolonged, frightening war-time situation. Secondly, the influence of context is manifested in the different roles as perceived by the person him- or herself.

The way in which the context is brought into the system has nothing to do with the real characteristics of that context, but rather reflects those relations with the context that are considered essential in this specific model. An important insight from dynamic systems theory is that living dynamic systems have no clear and fixed boundaries. The context cannot simply be defined as 'outside the skin' and the system as 'inside the skin'. It is at least partly a characteristic of the perspective of the observer: An individual may perceive unwanted 'inner voices' as being outside herself, outside her system, as 'not me'. Thoughts and emotions that are felt as unacceptable by the person may also be seen as 'context', or at least as 'not me'. A researcher, on the other hand, studying this person's conflicts, may regard these opposites as - conflicting - parts of a single system (e.g., Kunnen & Klein Wassink, 1999). In a dynamic systems approach one should, therefore, define which aspects belong to the system, which contextual factors can be considered important, and how these contextual factors may affect the system under study.

For the researcher who seeks to specify how the system is affected by the context, the three distinct conceptualizations of Dannefer and Perlmutter (1990) may be helpful. The influence of the context can be defined as (1) a random influence, (2) a factor having ordered effects, or (3) a system in itself. The differences between these conceptualizations become most explicit when the researcher wants to translate the relation with the context into a quantitative dynamic systems model. Therefore we will illustrate each conceptualization by describing such a 'translation'.

Context as **random influence** means that the self-organization of the system is affected in an unpredictable and non-systematic way from the outside. There is no systematic relation between subsequent disturbances or between the disturbances and the system. In a mathematical dynamic systems model, this could be expressed by adding or subtracting a small random number for each iteration. If context is seen as a factor that has **ordered effects**, there is a systematic relation between the contextual influences at different moments, and the context is more or less predictable. We already mentioned the example of living in a war-situation providing a long sequence of potentially frightening experiences. Context as an ordered effect can be represented in a model as a parameter. A lower or higher value of this parameter, then, expresses whether the context is more or less frightening. Such a parameter may differ between individuals and

situations, and change over time. If context is seen as a **system in itself**, it has its own internal dynamics, apart from the system under study. For example, the context could have its own stabilizing forces, which can counteract changes in the 'identity' system. In this case, a mathematical model could consist of two interacting systems or represent the context in the form of a separate logistic equation that is related to the other equations. Note that the system and the environment could equally well be seen as two interacting systems, or as one system. This ambiguity illustrates the point that systems have no intrinsically defined borders and environments.

Conceptualizing the context as a system has several advantages. First, it makes it clear that people are intrinsically related to their environment, while at the same time the environment is seen as being organized on its own terms, having its own dynamics and its own self-perpetuating tendencies and transitions. Secondly, the context can be meaningfully conceptualized at different levels of generality, in the same way that there are different levels of identity and self as shown in Table 9.1. Society at a macro-level concerns nations and institutions; on an intermediate level, schools, corporations, families etc.; and on a micro-level a face-to-face interaction between individuals (compare the discussion of Bronfenbrenner's work in chapter 2). The different levels of generality offer possibilities for a much more detailed and specific focus on the relation between identity and context. The conception of context as a system in itself represents the most thorough account of the relation between system and context. This characterizes the work of Fogel. He integrates the infant-system and the mother-system into one system that consists of the interaction between mother and infant, thereby including the dynamics of both the infant and the mother. However, this mother-plus-infant system also has a context. Which components are part of the system and which belong to the context therefore depends on the researcher's objectives.

A consequence of a self-organizational perspective is that, regardless of whether the context is conceptualized as random factor, ordered factor, or system, the effects of the contextual factor on the development of the system depend on its state. Random influences, for example, often have no effect at all. In an unstable system, a small random disturbance may cause a transition, however. In the same way, an ordered factor may have a positive effect, no effect, or a negative effect, again depending on the state of the system. If the context is seen as a system in itself, the

effects of the context not only depend on the target system, but the target system in turn affects the context. For example, whether an individual's self-esteem is affected by a critical remark strongly depends on that individual's appraisal. This appraisal results from the interaction between the identity-system and the context. The individual reaction, in turn, will affect the behavior of the person who made the remark, and thereby change the contextual system. Which may change the appraisals, and so on.

Assessing self and identity as self-organizing systems

The new and promising approach towards identity presented here will only be successful if it generates and is supported by empirical research. A final implication we therefore want to address is the operationalization, measurement, and analysis of key constructs.

The assessment of identity as a self-organizing process poses specific demands on the way information is measured and analyzed. Two notions of dynamic systems are especially relevant here. Firstly, self and identity are not structures that are always present, but rather phenomena that emerge in a self-organizing system. Secondly, essential characteristics of self and identity as a self-organizing system manifest themselves in the way they change over time.

A fundamental difference from previous conceptualizations is that stability of self and identity does not refer to structures that are always present. Stability means that comparable phenomena (EI's) emerge in different situations over time. We therefore have to find ways to 'grasp' these phenomena at the moment at which they are present. These ways may differ in accordance with the various types and aspects of self and identity that have been distinguished in this chapter. Implicit identity is the emerging system of self-referent EI's. The phenomena we want to observe, for example the emotional reactions when something concerning the person is at stake, only emerge in specific situations. This implies that the system should be studied when it is 'at work': in situations that matter with regard to self and identity. It cannot be assessed at any time and in any place, for example by administering a questionnaire. Several authors in this volume have developed different methods to assess the system 'at work'. Fogel and Verhofstadt-Denève observe their subjects directly in situations that matter to them. Fogel observes mothers and infants during their daily interactions, and Verhofstadt-Denève creates situations in psychodrama in which self-referent EI's are expected to emerge. Haviland-Jones et al., and Hermans and

Hermans-Jansen assess the working of the system more indirectly, by asking subjects about personally relevant roles and situations. By analyzing the relation between assessments, Haviland-Jones and colleagues and Hermans and Hermans-Jansen try to understand the structure of the higher-order system: the self or identity. Another possibility is to gather personal reports, such as diaries (Haviland-Jones et al.) and frequent open interviews (Kunnen & Klein Wassink, 1999), which yield relevant and emotional experiences in situations when the system was operating. In these methods, the researchers try to reveal non-conscious aspects of EI's by analyzing commonalities across situations.

Acquiring information about explicit identity is less difficult. Explicit identity is, by definition, the conscious and reflected narrative or identity. Although this identity need not always be present either, its emergence can be controlled by the person concerned. It can be assessed by asking people for their narratives, commitments, or simply 'who they are'. This is closest to the open type of traditional assessment methods of self and identity. This openness is a prerequisite here: one's stories and commitments are individual characteristics, and differ from person to person.

A sense of identity has been described as an all-encompassing feeling of sameness, continuity and uniqueness. This can be deducted from observations or assessed by direct questions concerning these feelings. Such questions have been developed, for example, by Blasi and Glodis (1995).

The second relevant notion is that a dynamic system is a system of related components that changes over time. A major difference between a dynamic systems, self-organizational approach and more traditional approaches is that the former approach studies phenomena from a diachronous perspective instead of a synchronous perspective. That is, it focuses on relations over time, instead of relations between variables (in subjects) at the same time. In the preceding chapters identity has been described as a time-related process that can only be studied from a diachronous perspective. Fogel, therefore, stresses that research should focus on sequences, in order to track how a psychological process unfolds over time in the context of co-regulated interactions. This implies that longitudinal observations are required for the study of self and identity. In addition, criteria should be formulated for the kinds of patterns of stability and

change that are relevant, and methods are needed to determine whether or not these patterns are present.

Most methods of analysis in the social sciences are directed towards assessing differences, rather than towards assessing change. Dynamic systems theory may help to identify the phenomena that are relevant in the search for change. The notion, for example, that a system often stabilizes in one of only a limited number of stable attractor states can be of help in the search for stable states. In a comparable way, dynamic systems theory describes specific types of behavioral characteristics that precede, accompany, and follow transitions in different types of systems. For example, an increased vulnerability for disruptions, and a longer return time (that means, a prolonged period of turbulence following disruptions) often precede a transition. These characteristics can be operationalized in observable behavior. Knowledge of bifurcation points - critical points at which development may follow either one pathway or the other - can also be helpful when observing the behavior of a system over time. Important questions are then: what triggers a bifurcation, and in what situations is such a transition more likely to occur? Because sudden changes may depend on gradual changes in underlying process parameters, such parameters should be found, and conditions should be defined under which bifurcations are more or less plausible.

Techniques that assess and analyze relevant dynamic systems behavior are still scarce. Lewis, Lamey and Douglas (1999) show how attractors and state space can be operationalized and demonstrated by simply registering well-defined behavior of the system over a certain period. Fogel focuses on interactions that have to do with mutual handling of toys and he demonstrates qualitative changes in this interaction. More traditional methods for analyzing change can also be very useful, however. Hermans and Hermans-Jansen compare assessments of an individual's valuations at different times, and Haviland-Jones and colleagues assess change by hierarchical cluster analysis to compare an adolescent's old roles with his or her new ones. Heymans stresses that some 'classical' methods of time-series analysis also offer possibilities to analyze change in systems. Often, these methods (Arima for example) appear to be more common in fields other than the social sciences. In addition, we point to a new form of factor analysis that can be used to assess change in individual time-series, developed by Molenaar (1985).

Another way to assess characteristics of a system is to build a quantitative model of the system, and to simulate its behavior. Such a model reflects the theoretical assumptions concerning the dynamics of the system. These assumptions can be tested by comparing the system's behavior with empirical phenomena or with other theoretical assumptions. To build such a model requires defining the components in the system (this step is needed in a qualitative approach as well), choosing the time scale, and turning the process of development into a few essential principles; moreover a of these components translation needs to be made into numbers, and of the process principles into equations. Haviland-Jones et al. give a clear example of this approach. They demonstrate how a few basic principles can be translated into a dynamic-systems model of the emergence of different emotions in a real time situation. Differences in personal and situational characteristics can be expressed in terms of the values of a few parameters. Relations between these characteristics and emerging outcomes can be used as a basis for the validation of the model. Other examples of the application of quantitative dynamic systems models in the field of self and identity are given in Kunnen, Bosma and Van Geert (in press), and in Kunnen and Bosma (2000).

Mathematical models can also be used to model the relations between micro-level systems. Emerging behavior on a lower level can be included as a parameter or variable in a higher-level system. Micro-level processes, then, may result in changes in parameters on a higher level of aggregation. Repeated frightening experiences – to follow the example of Haviland-Jones and colleagues - could result in an increased sensitivity to such experiences, and an increased chance that new situations will be experienced in a comparable way.

Most changes in identity will concern a macroscopic time scale, and can be assessed by comparing assessments over longer time periods. The duration of the therapies in the case studies of Hermans and Hermans-Jansen and Verhofstadt-Denève and the time covered by the diaries in Haviland-Jones et al., and the interview assessments of Kunnen and Klein Wassink (1999) range from months to years. Fogel observes mother-infant dyads over a shorter time span but here, too, the changes that take place within 'frames' cover a period of weeks or even months.

Studying and assessing self and identity as self-organizing systems require a thorough thinking and planning of assessment procedures. At this moment suitable techniques and

methods are not commonly available. However, more and more applications of dynamic systems views and techniques are appearing in many different fields. We mention just a few examples: social psychology (Nowak & Vallacher, 1999), language development (van Geert, 1991; Ruhland, 1998), cognitive psychology (van Geert, 1991; Eckstein, 1999; van der Maas & Molenaar, 1992), infant research (Fogel, Nwokah, & Karns, 1993), motor development (Thelen & Ulrich, 1991), psychopathology (Msihara, 1996), and personality (Svrakic, Svrakic, & Cloninger, 1996). Since (on a general level) the dynamic systems in these fields have much in common, the publications offer a wealth of inspiring and illustrative applications of dynamic systems.

The authors in the present volume offer elaborate and specific views on self and identity as emotional, dynamic systems, rooted in context. Despite clear difference of views, there is sufficient agreement to be able to formulate general as well as more specific characteristics of this new approach. The componential approach and the attention to time scales and aggregation levels provide new opportunities for the analysis of stability and change in the development of self and identity. As mentioned in our foreword, we see this book as a next step in a discussion and exploration of a new perspective in the field of self and identity. Not all the questions we posed at the beginning have been answered. Several still remain, some have disappeared, others have been reformulated. The question of how self and identity develop in adulthood, mentioned in Chapter 1, has, for instance, not been discussed at all. The conceptualization of self and identity as a self-organizing system, as worked out in this volume, however, does offer a framework, we hope, for a further exploration of these questions.